JUNWOO PARK

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EDUCATION

Sungkyunkwan University (SKKU) **Suwon, Korea**
B.S. in Computer Science and Engineering, GPA: 4.30/4.50 (3.94/4.00) Mar. 2024 – Feb. 2026
- Current Coursework: Computer Networks, Introduction to Software Engineering

Sejong University **Seoul, Korea**
Aerospace Engineering Mar. 2020 – Feb. 2024
- Fulfilled mandatory alternative military service (2022–2023) before transferring to SKKU.

RESEARCH EXPERIENCE

SKKU AI-Healthcare Lab **Suwon, Korea**
Research Assistant (Advisor: Prof. Sujee Lee) Feb. 2025 – Present
- Developed a multimodal deep learning model that integrates molecular graphs, sequences, images, and spectra using attention mechanisms for toxicity prediction [1].
- Collaborated with Samsung C-Lab on the "Muscle Tracker" project.

SKKU Human Language Intelligence Lab **Suwon, Korea**
Research Assistant (Advisor: Prof. JinYeong Bak) Sep. 2024 – Dec. 2024
- Analyzed and preprocessed user activity data including diary entries, therapeutic game logs, and media consumption, to characterize depression-related behavioral patterns.

PUBLICATIONS

[1] **J. Park** and S. Lee, MoltiTox: a multimodal fusion model for molecular toxicity prediction, *Frontiers in Toxicology*, accepted, 2025.

PROJECTS

Project SKKU Character: AI Platform for University Mascot, SKKU
AI Engineer Oct. 2025 – Present
- Secured 5M KRW (∼3,600 USD) in funding through the university entrepreneurship program.
- Developing an AI image generation system by fine-tuning a FLUX model on mascot datasets and implementing a FastAPI backend for generation requests.

Muscle Tracker: AI-driven Analysis for Strength Training, SKKU
Research Assistant Aug. 2025 – Nov. 2025
- Collected and preprocessed EMG and IMU sensor data from 14 subjects through a Python pipeline applying Butterworth filtering, interpolation, and RMS computation.
- Designed an algorithm for maximum muscle activation prediction using sigmoid curve fitting and contributed to exercise classification models, achieving 96% accuracy with a Temporal Convolutional Network.

NotiSKKU: Student Announcement App, SKKU
Team Leader Aug. 2024 – Nov. 2025
- Led a four-member team to develop a mobile app that aggregates campus announcements, based on surveys of over 60 students.
- Built the frontend using Flutter and launched the service on Google Play.

VisualVroom: Alert System for Hearing-Impaired Drivers, SKKU
Team Leader Apr. 2025 – Aug. 2025

- Directed a team to create a real-time auditory alert system for hearing-impaired drivers.
- Developed a Vision Transformer (ViT)-based model to classify audio source and direction from smartphone microphone input, achieving 83.9% overall accuracy.
- Integrated Google Speech-to-Text API to support in-car communication.

TEACHING & MENTORING EXPERIENCE

Course: Basis and Practice in Programming, SKKU
Teaching Assistant Fall 2025 – Present

- Guided 60 students in C programming coursework, evaluating assignments and providing feedback on code.

Software Mentoring Program, SKKU
Mentor May – Sep. 2024, 2025

- Taught Python fundamentals and conducted lab sessions for high school students.

EXTRACURRICULAR ACTIVITIES

Google Developer Groups on Campus, SKKU
President Feb. 2025 – Present

- Led ∼20 members, fostering collaboration through workshops and peer-to-peer projects.
- Established an international partnership with Mission San Jose High School in California, organizing programming competitions in Winter 2025 and a joint hackathon planned for Spring 2026.
- Hosted monthly tech talks featuring professionals and researchers from global tech companies.

AWARDS & HONORS

First Place, 2025 Uni-DTHON (Hackathon) Nov. 2025

Excellence Award, SW High School Creative Science-thon Nov. 2025

- Awarded for leading a team of 4 high school students as mentor.

2025 President's List, Sungkyunkwan University Sep. 2025

Grand Prize, ICT Award Korea 2025 Aug. 2025

SKILLS

Languages	Python, C, C++, Java, SQL, HTML/CSS
Frameworks & Tools	PyTorch, FastAPI, Flutter, Git, Docker, Linux, MATLAB